Original Article

Low Vision Intervention in Individuals with Age-related Macular Degeneration

Riddhi Ajay Bora, Priyanka Aher, Renu Ajay Magdum, Rohan Jagdish

Department of Ophthalmology, Dr. D. Y. Patil Medical College and Research Centre, Pune, Maharashtra, India

Abstract

Background: Age-related macular degeneration (AMD) is a leading cause of irreversible central vision loss in older adults. Low vision interventions (LVIs) aim to enhance residual visual function and promote independence. This study aimed to evaluate the clinical and functional outcomes of structured LVIs on visual performance, daily functioning, and quality of life in individuals with AMD. **Methods:** A prospective interventional study was conducted over 1 year at Dr. D. Y. Patil Medical College, Hospital, and Research Centre in Pune. Eighty-four participants with confirmed AMD and low vision were enrolled and assessed at baseline, 3 months, and 6 months postintervention. The rehabilitation program included optical, nonoptical, and electronic assistive devices, along with specialized visual training. Primary outcomes included best corrected visual acuity (BCVA), contrast sensitivity, and reading speed. Secondary outcomes were assessed using the National Eye Institute Visual Function Questionnaire-25 (NEI VFQ-25) and the modified Barthel index, evaluating self-reported visual function and independence in daily activities. **Results:** Of the 84 participants, 78 completed the full follow-up. Statistically significant improvements were observed in BCVA (mean gain: 0.17 logMAR), reading speed (mean increase: 18.8 words/min), and contrast sensitivity (mean gain: 0.22 units). The NEI VFQ-25 composite score improved by an average of 17.3 points, with marked enhancement in near-vision tasks and social engagement. Functional ability, as measured by the Modified Barthel Index, also showed meaningful improvement. No device-related complications were reported, and more than 85% of participants adhered to their prescribed rehabilitation plans. **Conclusions:** Structured low vision rehabilitation leads to significant improvements in visual performance, functional independence, and perceived quality of life among patients with AMD. These findings support the integration of LVIs into standard ophthalmic care protocols for elderly individuals with low vision.

Keywords: Age-related macular degeneration, assistive devices, low vision, quality of life, visual function, visual rehabilitation

Résumé

Contexte: La dégénérescence maculaire liée à l'âge (DMLA) constitue une cause majeure de perte irréversible de la vision centrale chez les personnes âgées. Les interventions de basse vision visent à optimiser la fonction visuelle résiduelle et à favoriser l'autonomie. Cette étude avait pour objectif d'évaluer les résultats cliniques et fonctionnels d'un programme structuré de réadaptation visuelle sur les performances visuelles, les activités quotidiennes et la qualité de vie chez des patients atteints de DMLA. **Méthodes:** Une étude prospective interventionnelle a été menée sur une période d'un an au Dr. D. Y. Patil Medical College, Hospital and Research Centre, à Pune. Quatre-vingt-quatre participants présentant une DMLA confirmée associée à une basse vision ont été inclus et évalués au départ, à 3 mois et à 6 mois après l'intervention. Le programme de réadaptation comprenait des aides optiques, non optiques et électroniques, ainsi qu'un entraînement visuel spécialisé. Les critères principaux comprenaient la meilleure acuité visuelle corrigée (MAVC), la sensibilité au contraste et la vitesse de lecture. Les critères secondaires incluaient le questionnaire NEI VFQ-25 et l'indice de Barthel modifié, permettant d'évaluer la fonction visuelle auto-rapportée et l'autonomie dans les activités de la vie quotidienne. **Résultats:** Parmi les 84 participants, 78 ont complété le suivi. Des améliorations statistiquement significatives ont été observées pour la MAVC (gain moyen : 0,17 logMAR), la vitesse de lecture (augmentation moyenne : 18,8 mots/min) et la sensibilité au contraste (gain moyen : 0,22 unité). Le score composite du NEI VFQ-25 s'est amélioré en moyenne de 17,3 points, avec des

Address for correspondence: Dr. Riddhi Ajay Bora, Flat No. 102 Sunit Laxmi Sharda Apartment, Ghole Road, Shivajinagar, Pune - 411 005, Maharashtra, India.
E-mail: riddhibora01@gmail.com

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progrès notables dans les activités de vision de près et l'engagement social. L'autonomie fonctionnelle, mesurée par l'indice de Barthel modifié, a également montré une amélioration significative. Aucune complication liée aux dispositifs n'a été signalée et plus de 85 % des participants ont respecté leur programme de réadaptation. **Conclusions:** La réadaptation visuelle structurée permet des améliorations significatives des performances visuelles, de l'autonomie fonctionnelle et de la qualité de vie perçue chez les patients atteints de DMLA. Ces résultats soutiennent l'intégration des interventions de basse vision dans les protocoles standards de soins ophtalmologiques chez les personnes âgées.

Mots-clés: Dégénérescence maculaire liée à l'âge, aides techniques, basse vision, qualité de vie, fonction visuelle, réadaptation visuelle

Introduction

Age-related macular degeneration (AMD) is one of the leading causes of vision loss in individuals aged 60 years and older. It is a chronic retinal disorder that significantly contributes to global visual impairment and disability.[1] The disease is primarily characterized by progressive degeneration of the macula, resulting in impaired central vision. This loss affects essential daily activities such as reading, driving, and facial recognition.

The global burden of AMD is rising steadily, largely due to increasing life expectancy. Current estimates suggest that the prevalence of AMD could exceed 288 million individuals by the year 2040.[1,2] In India, this burden is further compounded by demographic shifts and increased exposure to risk factors such as smoking, poor nutrition, and prolonged ultraviolet light exposure.[3]

AMD age-related macular degeneration is broadly classified into two forms: Dry (nonexudative) and wet (exudative). The dry form, which accounts for the majority of cases, typically progresses slowly and is associated with drusen accumulation and geographic atrophy. In contrast, the wet form is characterized by choroidal neovascularization and is associated with more rapid and severe vision loss.[4] While the advent of antivascular endothelial growth factor (anti-VEGF) therapy has significantly improved outcomes in wet AMD, effective treatments for dry AMD remain limited.[5] Consequently, there is a growing emphasis on low vision rehabilitation strategies to help patients preserve functional independence and make the most of their remaining vision. These interventions aim not only to enhance visual performance but also to improve the quality of life in affected individuals.

Low vision interventions (LVIs) are designed to optimize the use of residual vision through personalized strategies such as magnification devices, contrast-enhancing tools, orientation and mobility training, and environmental modifications.[6] Multiple studies have demonstrated that structured low vision rehabilitation significantly improves visual functioning, enhances psychological well-being, and reduces dependency in patients with AMD.[7,8]

However, the real-world implementation of LVIs, particularly in developing countries like India, is limited by several barriers. These include the lack of specialized rehabilitation services, inadequate referral systems, low patient awareness, and socioeconomic constraints.[9] Despite growing global evidence supporting the efficacy of LVIs, there remains a paucity of data from Indian tertiary care centers evaluating their impact on AMD populations.

A comprehensive understanding of patient-specific outcomes is crucial to inform service integration and guide policy development for vision rehabilitation. Therefore, the present study was undertaken at Dr. D. Y. Patil Medical College, Hospital and Research Center, Pimpri, Pune, over 1 year, with the objective of evaluating the clinical and functional outcomes of LVIs in individuals diagnosed with AMD. The primary aim was to assess improvements in visual task performance, autonomy in daily living, and overall patient satisfaction following tailored rehabilitation protocols administered in a structured clinical environment.

Methods

Study design and setting

This prospective interventional study was conducted from January to December 2024 by the Department of Ophthalmology at Dr. D. Y. Patil Medical College, Hospital and Research Center, Pimpri, Pune. Ethical clearance was obtained from the Institutional Ethics Committee prior to the initiation of the study. All participants provided written informed consent in accordance with the tenets of the Declaration of Helsinki.

Study population

A total of 84 participants were recruited from the outpatient ophthalmology clinic. All patients were clinically diagnosed with AMD and categorized as having low vision. Inclusion criteria were: Age ≥60 years, a documented history of chronic ocular disease for at least 3 months, and best corrected visual acuity (BCVA) in the better-seeing eye ranging from 6/18 to <3/60 (as per World Health Organization definition of low vision). Exclusion criteria included severe glaucoma, diabetic retinopathy, widespread cognitive impairment, and intraocular surgical procedure inside the preceding 3 months.

Baseline assessment

This included a thorough ocular examination, fundus exaination and the use of indirect ophthalmoscopy and spectral-area optical coherence tomography, the measurement of Best corrected visual acuity (BCVA) the usage of ETDRS charts, the assessment of contrast sensitivity the usage of the Pelli-Robson chart, and Amsler grid trying out.[10] The 15-object Geriatric Despair Scale is used to degree emotional and cognitive fame with the purpose to account for psychological factors which could affect practical outcomes.[11]

Low vision intervention protocol

Participants received a personalized low vision rehabilitation program tailored to their specific visual goals and functional limitations. The intervention comprised four key components.

Optical aids

Various optical devices were prescribed based on individual needs, including dome magnifiers, high-plus reading spectacles, handheld telescopes, and stand magnifiers.

Nonoptical aids

Supportive nonoptical tools were incorporated to improve task performance. These included high-intensity task lighting, reading stands, tactile markers, and large-print reading materials.

Electronic devices

Advanced electronic aids were introduced for patients with moderate to severe visual impairment. These included portable video magnifiers and closed-circuit television systems with customizable magnification and contrast settings to enhance visual accessibility.

Training modules

Each participant underwent a structured training program consisting of four supervised sessions spread over 8 weeks. The sessions were led by certified low vision rehabilitation professionals and focused on: These include the proper handling and use of prescribed low-vision devices, the implementation of eccentric-viewing strategies, and the development of adaptive techniques for performing activities of daily living (ALsD).

Outcome measures

The main effects of the have a look at have been upgrades in analyzing pace, contrast sensitivity, and best-corrected visual acuity (BCVA), as measured through Minnesota Low Vision Reading Test (MNREAD) charts 3 and 6 months after the intervention.[12] Functional independence and visual characteristics as stated by the patient, have been the primary and secondary goals. The National Eye Institute Visual Function Questionnaire-25 (NEI VFQ-25) was used to evaluate self-reported visual function.[13] The Modified Barthel Index was used to assess independence in activities of daily living (ADLs).[14]

Results

Participant characteristics

Out of the 84 individuals enrolled in the study, 78 participants (92.9%) successfully completed the 6-month follow-up. Attrition was due to relocation (n = 3), concurrent medical illness (n = 2), or voluntary withdrawal (n = 1). The cohort had a mean age of 68.4 ± 5.9 years (range: 61–81 years), with a slight male predominance (53.8%). The majority (67.9%) was retired, and 58.9% had attained at least secondary-level education. Clinically, 62.8% (n = 49) presented with dry AMD, while the remaining 37.2% (n = 29) had stable neovascular AMD managed with anti-VEGF therapy. Common systemic comorbidities included hypertension (42.3%), type 2 diabetes mellitus (34.6%), and hyperlipidemia (25.6%). The mean duration of AMD prior to enrollment was 3.1 ± 1.4 years. Detailed demographic and baseline clinical data are presented in Table 1.

Visual acuity and reading performance

Implementation of low vision rehabilitation protocols yielded notable improvements in visual outcomes. At baseline, the mean best-corrected visual acuity (BCVA) in the better-seeing eye was recorded as 0.82 ± 0.21 logMAR. Postintervention, a statistically significant improvement was observed, with BCVA improving to 0.72 ± 0.20 logMAR at 3 months and further to 0.65 ± 0.19 logMAR at 6 months ($P < 0.001$). These findings suggest enhanced visual clarity facilitated by tailored assistive devices and structured training sessions. While the improvements were more pronounced in cases with dry AMD, statistically meaningful gains were also documented in participants with stabilized wet AMD.

Reading performance also demonstrated considerable enhancement. The mean reading speed improved from a baseline of 45.3 ± 9.7 words per minute (wpm) to 57.8 ± 10.2 wpm at 3 months and 64.1 ± 11.5 wpm at 6 months ($P < 0.001$). These outcomes, derived from MNREAD chart assessments, reflect improved reading efficiency following consistent use of magnifiers and electronic visual aids. Corresponding data are detailed in Table 2.

Contrast sensitivity and central field function

Throughout the trial, there was a noticeable improvement in contrast sensitivity as determined by the Pelli-Robson chart. From 1.02 ± 0.18 at baseline to 1.14 ± 0.17 at 3 months and then to 1.24 ± 0.16 at 6 months, the mean score rose ($P = 0.002$).

Table 1: Baseline demographic and clinical characteristics

Parameter	Value
Number of participants	84
Mean age (years)	68.4±5.9
Gender (male/female)	45/39
AMD type (dry/wet) (%)	62.8/37.2
Mean duration of diagnosis (years)	3.1±1.4
Hypertension (%)	42.3
Diabetes mellitus (%)	34.6
Hyperlipidemia (%)	25.6
Education (≥ secondary)	58.9

AMD=Age-related macular degeneration

Table 2: Changes in objective visual outcomes

Outcome	Baseline	3 months	6 months	*P*
BCVA (logMAR)	0.82±0.21	0.72±0.20	0.65±0.19	<0.001
Reading speed (words per minute)	45.3±9.7	57.8±10.2	64.1±11.5	<0.001
Contrast sensitivity (Pelli-Robson)	1.02±0.18	1.14±0.17	1.24±0.16	0.002

BCVA=Best corrected visual acuity

The efficient use of contrast-enhancing visual aids and ideal lighting conditions was credited with these gains. By the end of 6 months, Amsler grid evaluations of participants with stable wet AMD revealed a discernible decrease in metamorphopsia severity in 61.5% of instances, indicating improved usage of peripheral retinal regions and increased central adaptability.

Subjective visual function and quality of life

The National Eye Institute Visual Function Questionnaire-25 (NEI VFQ-25) was used to document notable improvements in vision-related quality of life. At the 3-month mark, the composite score increased from 53.6 ± 8.4 to 63.2 ± 8.9, and at the 6-month mark, it reached 70.9 ± 9.3 ($P < 0.001$). The most significant improvements were found in near vision activities (mean increase: +19.5 points), social engagement (+13.2 points), and dependency reduction, according to subscale analysis. The wider effects of low vision rehabilitation on visual ability and psychological health are reflected in these gains [Table 3].

Independence in activities of daily living

Using the Modified Barthel Index, the mean independence score improved significantly from 68.5 ± 7.2 at baseline to 75.6 ± 7.0 at 3 months and 81.7 ± 6.8 at 6 months ($P < 0.001$). The largest improvements were observed in tasks requiring fine vision, such as grooming, dressing, and indoor mobility. At the end of the study, 64.1% of participants achieved scores indicating mild or no dependency, compared to 38.5% at baseline [Table 4].

Device utilization and adherence

At 6 months, 85.9% of patients reported regular use (≥4 days/week) of at least one low vision device. Among them, 72.4% used a combination of two or more aids (e.g., high-plus spectacles for near tasks and video magnifiers for extended reading). Electronic devices were most favored by participants under 70 years of age, while older patients preferred optical aids. Compliance was positively associated with higher education level and baseline VFQ-25 scores. No adverse effects or device-related injuries were reported. Only three participants (3.8%) discontinued device use due to difficulty in adaptation, despite repeated training sessions. Device-related feedback and satisfaction levels were high, particularly for mobility training and electronic magnification systems.

Table 3: National eye institute visual function questionnaire - 25 scores – Composite and subscale changes

Domain	Baseline	6 months	Mean change
Composite score	53.6±8.4	70.9±9.3	+17.3
Near activities	48.2±9.3	67.7±10.1	+19.5
Social functioning	52.1±7.6	65.3±8.8	+13.2
Dependency	50.3±6.9	61.9±7.2	+11.6
Role limitations	54.7±7.4	65.1±6.5	+10.4

Table 4: Functional independence (Modified Barthel Index)

Activity	Baseline	3 months	6 months	*P*
Overall score	68.5±7.2	75.6±7.0	81.7±6.8	<0.001
Grooming	62.4±8.1	70.2±7.9	77.6±7.5	<0.001
Dressing	65.3±7.8	72.8±7.6	79.3±7.0	<0.001
Mobility	67.5±6.9	74.3±7.1	80.9±6.6	<0.001

Discussion

The present study highlights the clinical relevance and measurable benefits of structured LVIs in patients with AMD, particularly in a resource-constrained, real-world Indian healthcare setting. The interventions led to statistically and clinically significant improvements in best corrected visual acuity (BCVA), reading speed, contrast sensitivity, and quality of life. These outcomes support the efficacy of personalized rehabilitation strategies and are consistent with findings reported in prior international studies.[7,8]

Beyond objective visual improvements, this study contributes to the growing body of evidence supporting the psychosocial and functional benefits of LVIs. The significant gains seen in both composite and subscale scores of the NEI VFQ-25 reflect not only improved visual performance but also reduced dependency and enhanced social engagement. These findings are consistent with those of Lamoureux *et al*., who demonstrated the positive impact of LVIs on psychosocial well-being and patient satisfaction.[7]

Marked enhancements in BCVA and reading speed underscore the value of device-based visual training. These observations are in line with the findings of Stelmack *et al.*, who reported significant improvements in visual function and performance in daily tasks among veterans undergoing similar rehabilitation protocols.[6]

The role of reading aids in improving functional vision and reading ability in individuals with low vision has also been well established in systematic reviews.[15] Similar benefits in reading rehabilitation using low vision devices have been reported among veterans.[16]

Our results also affirm that combining both optical and electronic assistive technologies offers a synergistic advantage, especially when tailored to each patient's cognitive and functional capacities.[16]

Reading speed, an important indicator of functional vision, was significantly improved in our study and is known to be reliably assessed using standardized tools such as the MNREAD test in patients with age-related macular degeneration.[17]

The improvements observed in contrast sensitivity, as assessed using the Pelli-Robson chart, further emphasize the role of appropriate lighting and magnification aids in optimizing visual function. Among individuals with stable neovascular AMD, a notable reduction in metamorphopsia symptoms was reported. This may be attributed to enhanced use of the peripheral retina and neuroplastic adaptations, mechanisms well-described in neuro-ophthalmic rehabilitation literature.[18]

Moreover, improvements in ADLs, as measured by the Modified Barthel Index, highlight the practical value of these interventions in enhancing patient autonomy and reducing caregiver burden–an aspect often under-recognized in conventional AMD management strategies.

Despite the encouraging results, certain limitations must be acknowledged. The absence of a control group limits the ability to draw definitive causal inferences, underscoring the need for future randomized controlled trials to validate these findings. In addition, the reliance on self-reported data for device usage introduces the potential for reporting bias. Nevertheless, within these limitations, the study provides valuable insight into the feasibility, effectiveness, and clinical applicability of structured low vision rehabilitation programs in tertiary care settings across India.

Conclusiion

Structured low vision rehabilitation leads to significant improvements in visual performance, functional independence, and perceived quality of life among patients with AMD. These findings support the integration of low vision interventions into standard ophthalmic care protocols for elderly individuals with low vision.

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Nil.

Conflicts of interest

There are no conflicts of interest.

References

1. Wong WL, Su X, Li X, Cheung CM, Klein R, Cheng CY, *et al.* Global prevalence of age-related macular degeneration and disease burden projection for 2020 and 2040: A systematic review and meta-analysis. Lancet Glob Health 2014;2:e106-16.
2. Mitchell P, Liew G, Gopinath B, Wong TY. Age-related macular degeneration. Lancet 2018;392:1147-59.
3. Klein R, Klein BEK, Linton KLP. Prevalence of age-related maculopathy: The Beaver Dam Eye Study. Ophthalmology. 1992;99:933-43.
4. Ferris FL 3rd, Wilkinson CP, Bird A, Chakravarthy U, Chew E, Csaky K, *et al.* Clinical classification of age-related macular degeneration. Ophthalmology 2013;120:844-51.
5. Cheung LK, Eaton A. Age-related macular degeneration. Pharmacotherapy 2013;33:838-55.
6. Stelmack JA, Tang XC, Reda DJ, Rinne S, Mancil RM, Massof RW, *et al.* Outcomes of the veterans affairs low vision intervention trial (LOVIT). Arch Ophthalmol 2008;126:608-17.
7. Lamoureux EL, Pallant JF, Pesudovs K, Rees G, Hassell JB, Keeffe JE. The effectiveness of low-vision rehabilitation on participation in daily living and quality of life. Invest Ophthalmol Vis Sci 2007;48:1476-82.
8. Goodrich GL, Kirby J, Cardenas-Dow T, Fine FM. Training and low vision devices in rehabilitation of patients with age-related macular degeneration. J Vis Impair Blind 2002;96:663-72.
9. Murthy GV, Gupta SK, Bachani D, Jose R, John N. Current estimates of blindness in India. Br J Ophthalmol 2005;89:257-60.
10. Hee MR, Puliafito CA, Wong C, Duker JS, Reichel E, Schuman JS, *et al.* Optical coherence tomography of macular holes. Ophthalmology 1995;102:748-56.
11. Almeida OP, Almeida SA. Short versions of the geriatric depression scale: A study of their validity for the diagnosis of a major depressive episode according to ICD-10 and DSM-IV. Int J Geriatr Psychiatry 1999;14:858-65.
12. Virgili G, Acosta R, Bentley SA, Giacomelli G, Allcock C, Evans JR. Reading aids for adults with low vision. Cochrane Database Syst Rev 2018;4:CD003303.
13. Mangione CM, Lee PP, Pitts J, Gutierrez P, Berry S, Hays RD. Psychometric properties of the National Eye Institute Visual Function Questionnaire (NEI-VFQ). NEI-VFQ field test investigators. Arch Ophthalmol 1998;116:1496-504.
14. Shah S, Vanclay F, Cooper B. Improving the sensitivity of the Barthel Index for stroke rehabilitation. J Clin Epidemiol 1989;42:703-9.
15. Watson GR, De l'Aune W, Long S, Maino J, Stelmack J, Sovich B. Veterans' use of low vision devices for reading rehabilitation: A pilot study. J Vis Impair Blind 1997;91:505-15.
16. Virgili G, Rubin G. Orientation and mobility training for adults with low vision. Cochrane Database Syst Rev 2010;2010:CD003925.
17. Subhi Y, Kjer B, Munch IC. Precision of reading speed measurements with the MNREAD test in patients with age-related macular degeneration. Acta Ophthalmol 2020;98:e638-44.
18. Plank T, Frolo J, Brandl-Rühle S, Renner AB, Helbig H, Greenlee MW, *et al.* Functional reorganization of the visual cortex in patients with age-related macular degeneration. Invest Ophthalmol Vis Sci 2011;52:6073-80.